Assignment 1

Suggested Solutions

Q1. Compute the empirical and generator-based estimators of one-year transition matrix for the following data.

(a) There are four credit classes, A, B, C, D. At the beginning, there are 20 firms in A, B, C, each.

(b) In one month, two B rated firms migrate to C, and two B rated firms migrate to A , and these firms stay there for the rest of the year.

(c) In three months, three A rated firm migrates to B, and stay there for the rest of the year.

(d) In seven months, two C rated firms migrate to B and stay there for the rest of the year. Furthermore, one C rate firm defaults.

(e) Ten months later, two C rated firms default.

Solution.

$$\hat{P}_{empirical} = \begin{pmatrix} \frac{17}{20} & \frac{3}{20} & 0 & 0 \\ \frac{2}{20} & \frac{16}{20} & \frac{2}{20} & 0 \\ 0 & \frac{2}{20} & \frac{15}{20} & \frac{3}{20} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We compute generator-based estimators as follows,

$$N_A = \frac{235}{12}, \qquad N_B = \frac{233}{12}, \qquad N_C = \frac{81}{4}$$

$$M_{A,B} = 3, \qquad M_{A,C} = M_{A,D} = 0$$

$$M_{B,A} = 2, \qquad M_{B,C} = 2, \qquad M_{B,D} = 0$$

$$M_{C,A} = 0, \qquad M_{C,B} = 2, \qquad M_{C,D} = 3$$

The estimator becomes

$$\hat{\lambda}_{A,B} = \frac{M_{A,B}}{N_A} = \frac{36}{235}, \qquad \hat{\lambda}_{A,C} = \hat{\lambda}_{A,D} = 0$$

$$\hat{\lambda}_{B,A} = \frac{M_{B,A}}{N_B} = \frac{24}{233}, \qquad \hat{\lambda}_{B,C} = \frac{M_{B,C}}{N_B} = \frac{24}{233}, \qquad \hat{\lambda}_{B,D} = 0$$

$$\hat{\lambda}_{C,B} = \frac{M_{C,B}}{N_C} = \frac{8}{81}, \qquad \hat{\lambda}_{C,D} = \frac{M_{C,D}}{N_C} = \frac{4}{27}, \qquad \hat{\lambda}_{C,A} = 0$$

The estimator of the generator matrix is given by

$$\hat{\Lambda} = \begin{pmatrix} -\frac{36}{235} & \frac{36}{235} & 0 & 0 \\ \frac{24}{233} & -\frac{48}{233} & \frac{24}{233} & 0 \\ 0 & \frac{8}{81} & -\frac{20}{81} & \frac{4}{27} \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

Therefore the one-year transition matrix is estimated by

$$\hat{P}_{generator} = e^{\hat{\Lambda}}$$

$$= \begin{pmatrix} 0.864631 & 0.128572 & 0.006462 & 0.000336 \\ 0.086450 & 0.824468 & 0.082497 & 0.006584 \\ 0.004166 & 0.079102 & 0.785246 & 0.131486 \\ 0.000000 & 0.000000 & 0.000000 & 1.000000 \end{pmatrix}$$

Q2. A company has issued 3 - and 5 -year bonds with a coupon payment of 4.1% per year payable annually. The yields on the bonds (in terms of continuous compounding) are 4.6% and 4.75%, respectively. Assume the risk-free rate is 2.5% for all maturities, and the recovery value is 40% of the par value. Also assume default can only take place halfway through each year, i.e. at year $0.5, 1.5, 2.5$, etc. Compute P_1 and P_2, which are the implied default intensities per year for years 1 to 3 and for years 4 and 5, respectively.

Solution.

For 3 years:

Time (Years)	Default Intensity (per year)	Recovery Value ($)	Risk-Free Value ($)	Loss Given Default ($)	Discount Factor	PV of Expected Loss ($)
0.5	P_1	40	105.7911	65.7911	0.9876	$64.9738P_1$
1.5	P_1	40	104.3176	64.3176	0.9632	$61.9504P_1$
2.5	P_1	40	102.8068	62.8068	0.9394	$59.0016P_1$
					Total:	$185.9257P_1$

Since the yield is 4.6% the corporate bond price is

$$4.1e^{-0.046*1} + 4.1e^{-0.046*2} + 104.1e^{-0.046*3} = 98.3367$$

Similarly, the default-free bond price is

$$4.1e^{-0.025*1} + 4.1e^{-0.025*2} + 104.1e^{-0.025*3} = 104.4769$$

Setting the difference between the corporate bond and default-free bond equal to the PV of the expected loss yields

$$101.2257 - 98.3462 = 185.9257P_1,$$

i.e. the implied default intensity (or implied one-year default probability given no default yet) is

$$P_1 = 3.3025\%$$

For 5 years:

Time (Years)	Default Intensity (per year)	Recovery Value ($)	Risk-Free Value ($)	Loss Given Default ($)	Discount Factor	PV of Expected Loss ($)
0.5	P_1	40	108.6297	68.6297	0.9876	$67.7772P_1$
1.5	P_1	40	107.2281	67.2281	0.9632	$64.7538P_1$
2.5	P_1	40	105.7911	65.7911	0.9394	$61.8050P_1$
3.5	P_2	40	104.3176	64.3176	0.9162	$58.9290P_2$
4.5	P_2	40	102.8068	62.8068	0.8936	$56.1240P_2$
					Total:	$194.3360P1$ $+115.0530P2$

Now since the yield is 4.75% the corporate bond price is

$$4.1e^{-0.0475*1} + 4.1e^{-0.0475*2} + 4.1e^{-0.0475*3} + 4.1e^{-0.0475*4} + 104.1e^{-0.0475*5} = 96.6772$$

Similarly, the default-free bond price is

$$4.1e^{-0.025*1} + 4.1e^{-0.025*2} + 4.1e^{-0.025*3} + 4.1e^{-0.025*4} + 104.1e^{-0.025*5} = 107.2803$$

Setting the difference between the corporate bond and default-free bond equal to the PV of the expected loss yields

$$107.2803 - 96.6772 = 194.3360P1 + 115.0530P2,$$

i.e. the implied default intensity (or implied one-year default probability given no default yet) is

$$P_2 = 3.6376\%$$